THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** lustre	metallic lustre
2	**(d)** Mercury	used in some thermometers
3	**(a)** iodine	shiny greyish-black solid
4	**(c)** rust	iron oxide
5	**(b)** tarnish	reaction with sulfur compounds
6	**(b)** basic	turn red litmus blue
7	**(b)** ductile and a good conductor	properties
8	**(c)** carbon	fuel
9	**(c)** Nichrome (alloy)	high resistance, melting point
10	**(a)** mixture of metals (or a metal and non-metal)	e.g. steel, brass

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Malleability: beaten into sheets, e.g. aluminium foil (1); ductility: drawn into wires, e.g. copper (1).	1+1
12	The shine of a surface (1); gold and silver (1).	1+1
13	Mercury and bromine (2).	2
14	Producing a ringing sound on being struck (1); a bell (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Heat one end of a metal rod with wax beads (1); beads melt and fall in order (1); heat flows through metal (1).	1+1+1
16	Metals lustrous, hard, conduct (1.5); non-metals dull, brittle, poor conductors (1.5).	1½+1½
17	Iron reacts with oxygen and moisture to form rust (1.5); painting, galvanising (1.5).	1½+1½
	OR: Three test tubes: nail in air and water, in boiled water with oil, in dry air with calcium chloride (1.5); only the first rusts (1.5).	1½+1½
18	Copper gets a green coating (1.5); silver turns black (1.5).	1½+1½
	OR: It is malleable (1); does not corrode easily (1); is non-toxic (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Lustre, hardness, malleability, ductility, sonority/conduction (2). (ii) Definitions (2). (iii) Uses (1).	2+2+1
	OR: (i) Iron reacting with air and water (1). (ii) Corrosion (1). (iii) Oil, paint, galvanisation (1). (iv) Alloy resists corrosion (2).	1+1+1+2
20	(i) Three set-ups (2). (ii) Rust only in air plus water (2). (iii) Both needed (1).	2+2+1
	OR: (i) Gold is less reactive (1). (ii) A layer formed by gases/air (1). (iii) Clean with polish (1). (iv) Lustrous and non-corroding (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** malleability	e.g. aluminium foil
2	**(c)** Bromine	reddish-brown liquid
3	**(a)** Sulfur	breaks into pieces when hit
4	**(a)** both oxygen (air) and water	iron, air and moisture
5	**(c)** galvanisation	e.g. buckets
6	**(b)** acidic	turn blue litmus red
7	**(d)** aluminium	malleable
8	**(b)** iron	haemoglobin
9	**(d)** corrosion	protects iron
10	**(c)** carbon	stronger than iron

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Dull appearance, brittle, poor conductors (2).	2
12	Formation of a brown layer on iron (1); air and water (1).	1+1
13	Painting; galvanisation; oiling (2).	2
14	Gradual damage of metal surfaces by air, water (1); rusting and green coating on copper (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Oxygen for respiration, carbon as fuel, iodine as antiseptic, sulfur in matches (3).	3
16	Sonorous (1.5); produce a clear ringing sound (1.5).	1½+1½
17	Coating iron with zinc (1.5); zinc blocks air and water and corrodes first (1.5).	1½+1½
	OR: Powder: non-metal, brittle (1.5); flattens: metal, malleable (1.5).	1½+1½
18	It conducts electricity (1.5); it is soft and leaves a mark (1.5).	1½+1½
	OR: Mercury is a liquid metal; sodium is soft; graphite conducts; iodine is lustrous (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Table of 4 properties (2). (ii) Mercury, graphite, iodine (2). (iii) Show rules have exceptions (1).	2+2+1
	OR: (i) Mg burns, copper turns green, iron rusts (2). (ii) Oxide products (2). (iii) Magnesium (1).	2+2+1
20	(i) Paint wears off (1). (ii) Rust weakens it (1). (iii) Galvanisation (1). (iv) Stainless steel (2).	1+1+1+2
	OR: (i) Needed for life and industry (1). (ii) Oxygen, nitrogen, sulfur, iodine (2). (iii) Forms diamond and graphite (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.